

Answer all the questions

1. What is the distance between the first bright fringe and the central maximum on the screen when the wavelength of the light from the source is 546 nm and the slits separation is 0.12 mm. The screen is 60 cm in front of the slits.
[2.73 mm]

2. In an experiment, a student projected a monochromatic light towards the slits of separation 0.05 mm. The bright and dark fringes pattern can be seen on a screen which is situated at 75 cm from the slit. The distance between the seven order bright fringes below the central is 12 cm. Calculate the wavelength of the light fringe that used.
[5.71 x 10⁻⁷m]

3. Monochromatic light illuminates a double-slit system with a slit separation 0.3 mm. The second-order maximum occurs at $y = 4.0$ mm on a screen 1.0 m from the slit. Find,
(a) the wavelength,
(b) the distance on the screen between the central maximum and the third order-minimum.
[6 x10⁻⁷ m, 5.0 x 10⁻³ m]

4. In a Young's double-slits experiment, the separation between second and third bright fringe is 6.25×10^{-4} m when the wavelength used is 6.2×10^{-7} m. The distance from the slits to the screen is 0.8 m. Calculate the separation of the two slits.
[7.94 x 10⁻⁴ m]

5. In a Young's double-slit experiment, the slit separation is 0.2 mm and a screen is 0.4 m from the double-slit. A light source that emit red and violet light of wavelengths 700 nm and 420 nm is used. Calculate
(a) the fringe separation between red fringes
(b) the fringe separation between violet fringes,
[1.4 x 10⁻³m, 0.84 x 10⁻³m]

6. The separation of a double-slit is 0.24 mm. A beam of monochromatic light of wavelength 600 nm is incident on this double-slit. If the screen is located 1.0 m away from the slit. Calculate the distance from the central bright fringe to the
(a) third bright fringe
(b) third dark fringe
[7.5 x 10⁻³m, 6.25 x 10⁻³m]

7. With two slits spaced 0.2 mm apart and a screen at a distance of 1.0 m, the fifth bright fringe is found to be displaced at 7.5 mm from the central fringe. Find the wavelength of the light used.
[3.0 x 10⁻⁷m]

8. In a Young's double slit experiment, the fringe pattern contains 5 complete fringes in a distance of 4.5 mm. Calculate the wavelength of the light if the ratio of the distance from the slits to the screen D , to the slit separation d is 500.
- $[6.0 \times 10^{-7}\text{m}]$**
9. In a Young's double slit experiment, the separation of the slits is 0.2 mm. The screen is 1.5 m away from the slits. The separation between the two 5th dark fringes on both sides of the central fringe is 4.05 cm. Determine the wavelength of the light used.
- $[6.01 \times 10^{-7}\text{m}]$**
10. Light of wavelength 600 nm is incident upon a Young's double slits which are 1.0 m from a screen. The slit separation is 0.3 mm. Determine the separation between the 2nd bright fringe and the 4th dark fringe.
- $[3 \times 10^{-3}\text{m}]$**